

relval - Release Validation

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2026/05/18

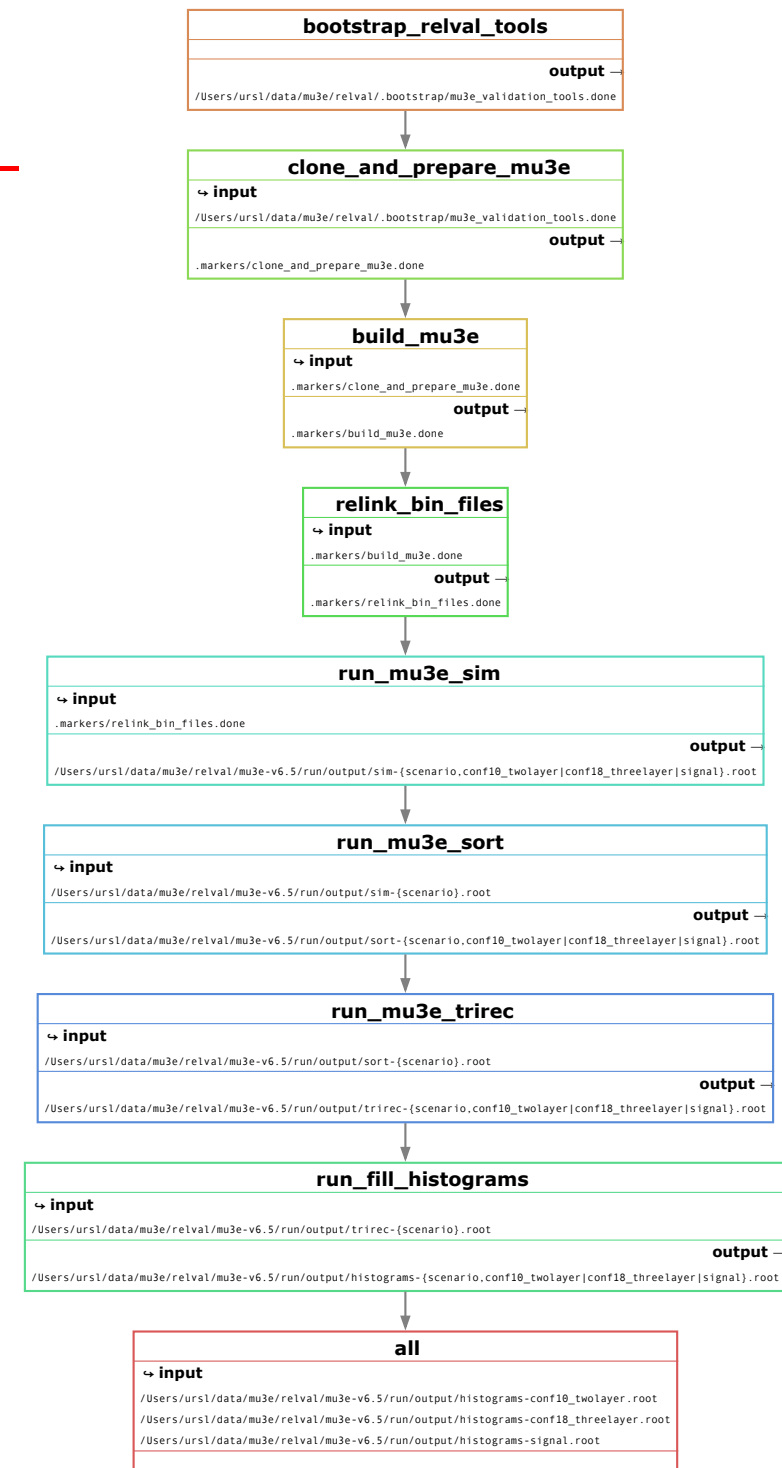
- Introduction/Motivation
- Setup
- Discussion

What and why "Release Validation"?

- **Standard** set of workflows on **simulated (MC)** and **detector** datasets
 - ▷ tests the entire pipeline
 - event generation
 - simulation (Geant4)/digi
 - reconstruction
 - CDB GT/payloads/etc
- **Performance** comparisons
 - ▷ physics performance of new code releases(/tags)
 - ▷ verifies technical adaptations (coding/algorithms/. . .)
 - ▷ CDB GT/payloads/etc - MC scenarios(?)
- **Why this talk?**
 - ▷ I want to validate CDB code and contents
 - ▷ I want to experiment with workflow management systems
 - in view of the **prompt workflow**, non-trivial setup - possibly useful for collaboration
 - ▷ I am not aware of others' validations of **vX.Ypre0**
 - ▷ **release validation is part of DQM!**

Setup

- snakemake [setup](#)
- Initialization
 - ▷ Bootstrap (if necessary)
 - ▷ checkout mu3e_tag (and submodules)
 - ▷ build, symlink huge binary files
- Run mu3e programs
 - ▷ mu3eSim for various scenarios (signal, conf10, conf18, . . .)
 - ▷ mu3eSort, with CDB
 - ▷ mu3eTriRec
- Analysis part
 - ▷ (dedicated) histogram filling (CDB, . . .)
 - ▷ RunCompare PDF summaries
 - ▷ mu3eTreeDumper, Histocompare
- Publication <https://mu3edb0/relval>



Comments

- Invocation rather trivial

```
./runRelval --all
./runRelval -j4 -p --compare-against-setup relval_mu3e-v6.6_mcidealv6.5 \
    --config mu3e_tag=v6.7pre0 cdb_GT=mcidealv6.5
```

- Can rely on packaged software, e.g.,

```
podman run --rm \
    -v "/Users/ursl/data/mu3e/relval:/relval:ro" \
    -v "/Users/ursl/data/mu3e/relval/mu3e-relval_mu3e-v6.4.4_mcidealv6.5/run/output/compare/conf1" \
    docker.io/mu3e/histocompare \
    "$dump1_data_obj" \
    "$dump2_data_obj" \
    --treedump --threshold 0.60 --wasserstein 0.60 --accFailFraction 0.05 \
    --pdf -o "$out_name_obj" --skip=eventWeight --skip=farm_status \
    --skip=frameID --skip=runID --skip=mc_eventID --skip=mc_weight \
    --twoDimThreshold 1.0
```

▷ run versioned/tagged histoCompare executables in podman?

- Platform independent

- ▷ BSD UNIX - MBP
- ▷ Ubuntu - mu3edb0 (completely under our control)
- ▷ merlin?? (maybe **prompt** will run on mu3edb0, submitting to merlin6 batch?)

- Small node.js server (next to CDB/RDB) for validation results

▷ <https://mu3edb0.psi.ch/relval>

My focus: CDB

- I will continue with relval with CDB focus
- Technicalities
 - ▷ compiler warnings
 - ▷ (new) GT well defined with all included tags
 - ▷ payload classes, payload validation, etc
 - mu3eTrirec (etc) run successfully to completion
- Payload performance
 - ▷ resolutions, efficiencies, . . .
 - ▷ defective components, e.g., with new GTs
- Samples?
 - ▷ signal MC (mcideal)
 - ▷ conf10 (mcrealistic)
 - ▷ conf18 (mcrealistic)
 - ▷ data? (likely I should)
- Much still to be implemented

Quo vadis? Discussion

- relval is not quite the same as CI or MC validation
- So far basically the skeleton
 - ▷ technical setup
 - ▷ some (still developing) histogramming/comparison
 - ▷ relval does not care what exactly is run
 - it can run anything anybody wants
- Apart from CDB validation, to be continued?
 - ▷ how? In what setting?
 - ▷ deliverables?
 - pdfs
 - parsing of log files (histogram comparisons test scores, . . .)
 - datasets: "signal"? MC configurations?
 - ▷ To be really useful in wider context, more should be implemented
 - histoCompare: somebody else! Ideally subsystems - "physics" - DQM

⇒ Look at outcome and react!?